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CSES Problem Set**Hidden Integer**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a hidden integer x . Your task is to find the value of x .

To do this, you can ask questions: you can choose an integer y and you will be told if $y < x$.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You can start asking questions right away.

On your turn, you can print one of the following:

- "? y ", where $1 \leq y \leq 10^9$: ask if $y < x$. The grader will return YES if $y < x$ and NO otherwise.
- "! x ": report that the hidden integer is x . Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $1 \leq x \leq 10^9$
- you can ask at most 30 questions of type ?

Example

```
? 3
YES
? 6
YES
? 7
NO
! 7
```

CSES Problem Set

Hidden Permutation

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a hidden permutation $a_1, a_2, \dots, a_n$ of integers $1, 2, \dots, n$. Your task is to find this permutation.

To do this, you can ask questions: you can choose two indices i and j and you will be told if $a_i < a_j$.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n : the length of the permutation.

On your turn, you can print one of the following:

- `"? i j"`, where $1 \leq i, j \leq n$: ask if $a_i < a_j$. The grader will return YES if $a_i < a_j$ and NO otherwise.
- `"! a1 a2 ... an"`: report that the hidden permutation is $a_1, a_2, \dots, a_n$. Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $1 \leq n \leq 1000$
- you can ask at most 10^4 questions of type ?

Example

```
3
? 3 2
NO
? 3 1
YES
! 3 1 2
```

Explanation: The hidden permutation is $[3, 1, 2]$. The first question asks if $a_3 < a_2$ which is false, so the answer is NO. The second question asks if $a_3 < a_1$ which is true, so the answer is YES.

CSES Problem Set

Permuted Binary Strings

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a hidden permutation $a_1, a_2, \dots, a_n$ of integers $1, 2, \dots, n$. Your task is to find this permutation.

To do this, you can ask questions: you can choose a binary string $b_1 b_2 \dots b_n$ and you will receive the binary string $b_{a_1} b_{a_2} \dots b_{a_n}$.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n : the length of the permutation.

On your turn, you can print one of the following:

- "? $b_1 b_2 \dots b_n$ ", where $b_i \in \{0, 1\}$: The grader will return the binary string $b_{a_1} b_{a_2} \dots b_{a_n}$.
- "! $a_1 a_2 \dots a_n$ ": report that the hidden permutation is $a_1, a_2, \dots, a_n$. Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $1 \leq n \leq 1000$
- you can ask at most 10 questions of type ?

Example

```
3
? 100
100
? 010
001
? 001
010
! 1 3 2
```

Explanation: The hidden permutation is $[1, 3, 2]$. In the first question $b_1 b_2 b_3 = 100$ and the grader returns $b_{a_1} b_{a_2} b_{a_3} = b_1 b_3 b_2 = 100$. In the second question $b_1 b_2 b_3 = 010$ and the grader returns $b_1 b_3 b_2 = 001$.

CSES Problem Set

Colored Chairs

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n chairs arranged in a circle. Each chair is either red or blue. The chairs are numbered $1, 2, \dots, n$; chairs i and $i + 1$ are next to each other for all $1 \leq i \leq n$. Here chair $n + 1$ refers to chair 1.

Your task is to find two chairs that have the same color and are next to each other.

To do this, you can ask questions: you can choose a chair and you will be told the color of that chair.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n : the number of chairs.

On your turn, you can print one of the following:

- " $? i$ ", where $1 \leq i \leq n$: ask the color of chair i . The grader will return R or B for red or blue.
- " $! i$ ": report that chairs i and $i + 1$ have the same color. Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $3 \leq n \leq 2 \cdot 10^5$, n is odd
- you can ask at most 20 questions of type $?$

Example

```
5
? 1
R
? 2
B
? 3
B
! 2
```